FastLane: A Voice Agent That Uses a Small Model for Easy Turns and a Large Model Only for Hard Ones

LMA Group Project 2026 — Mid-Submission Report (30 October 2026)

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GitHub: `https://github.com/Sankalp0109/FastLane-Adaptive-Voice-Agent`

Abstract

FastLane is a phone agent that books, changes and cancels movie tickets, bus tickets and restaurant tables for Indian-English callers. A LoRA-tuned Qwen2.5-3B answers every turn first and passes only hard turns to a large API model (gpt-oss-120b) or to staff; every action goes through a tool and policy gateway that enforces booking rules and a read-back confirmation in code. This report describes everything built so far: a deterministic practice booking system with 11 tools, a 360-call test set with a simulated caller, a data generator that runs an oracle agent through the real orchestrator, five data versions and five Kaggle training sessions, and a local voice pipeline (Silero VAD, Whisper, Kokoro). Offline, the tuned small model routes 99.3% of held-out steps correctly (untuned: 89.1%) and fills tool arguments exactly in 99.0% (untuned: 26.8%). On 24 end-to-end calls, FastLane with the fixes found in the pilots succeeds in 95.8% of calls (untuned small model: 79.2%; untuned small model alone, B2: 29.2%) with the lowest median model time per turn (1.55 s). The ablations support two hypotheses: without date and alternative tools (A2) the large model is called about twice as often, and with rules only in the prompt (A3) 15 of 24 calls make a booking change the caller never confirmed, against none for FastLane. The large-model-only baseline (B1) could not be run before this report because the free API quota ran out. On real Indian-English speech (Svarah), Sarvam's recogniser has the lowest error (5.8% WER vs. 9.5–10.5% for Whisper), and a TTS→STT round trip shows that recogniser noise roughly triples the tuned model's argument errors (98.1% → 94.3% exact). We report each iteration with the reason it was needed, and what remains for the final comparison.

1 Introduction

A voice agent has about a second to reply before callers talk over it or repeat themselves. Most agents send every turn to a large API model, which adds 1–3 s per turn and costs too much for a small cinema, bus operator or restaurant. Yet most turns in a booking call are easy (a date, a seat count, a name, a "yes" to a read-back); only a few are hard ("next Friday", a sold-out show, a caller changing their mind, a request that breaks a rule). A wrong booking is worse than a slow one, and recognisers often mishear Indian names, places and numbers. Our question is *for each turn, which model should reply*, judged on speed, correctness and safety together.

The proposal states three research questions: (RQ1) can a LoRA-tuned small model that both answers easy turns and decides when to pass a turn on cut time per turn by $\geq$50% without a real drop in booking success compared with a large-model-only agent; (RQ2) does moving rules and routine work (dates, alternatives, policy checks) from the prompt into code reduce large-model calls and unsafe bookings; (RQ3) does a small model trained only on movies and buses still work on restaurants. They come with seven hypotheses (Table 11).

Progress and structure of this report. Table 1 summarises progress by phase of the plan: Phases 0–7 are done, and Phase 8 (the full comparison) has started with baseline and ablation runs on a 24-call sample.

Phase	Status	What exists
0 Scaffolding	done	Python package, CLI (`fastlane ...`), 192 tests
1 Practice world	done	SQLite booking system for 3 domains, fixed seed, snapshot/restore per call
2 Gateway and booking record	done	11 tools, rules in code, read-back gate, identity check, error injection, 2 rules added after pilots
3 Retrieval	done (corpus partial)	BM25 over 25 policy documents, old/new policy versions, early lookup; Recall@3 0.967
4 Models and orchestrator	done	all 12 systems run end to end; large model with rate-limit backup
5 Evaluation harness	done	360-call test set, LLM caller, resumable runner, all proposal metrics
6 Training	v1–v3 trained; v4/v5 data ready	5 data versions, 5 Kaggle sessions; v3: route 99.3%, arguments 99.0% offline
7 Speech	done	browser voice demo, STT comparison on Svarah, TTS→STT noise for training
8 Experiments	started	FastLane, B2, A2, A3, A8 on 24 calls (A1 from the pilot); B1, B3 blocked by API quota

Table 1: Progress by phase of the project plan.

The report follows the mid-submission items: the system (Section 3), datasets (Section 4), baselines, ablations and metrics (Section 5), training (Section 6), results of all experiments run so far (Section 7), challenges and mitigation (Section 8), the status of each hypothesis (Section 9) and the remaining work with a timeline to the final submission on 15 November (Section 10). All code is in one Python package (about 10.4k lines in `src/fastlane`, 192 passing tests) on GitHub, with the Kaggle notebook, result files and a run log recording every experiment.

2 Related Work

Model routers such as FrugalGPT (a cascade of LLMs of increasing cost) and RouteLLM (a router trained on preference data) choose one model per *question* on single-turn benchmarks. FastLane routes *per turn* inside a long, tool-using dialogue, and the router is the small model itself, so easy turns pay no routing delay. Tool-agent benchmarks such as τ-bench check the final database state with a simulated user, which we adopt, but do not measure latency, so they reward always using the largest model; we report speed, success and unsafe actions together. We tune Qwen2.5-3B-Instruct with LoRA on a 4-bit base (QLoRA) and serve it with MLX; the speech layer uses Silero VAD, Whisper, Sarvam's Indian-language recogniser and Kokoro TTS.

3 System

Practice world. One SQLite database built from a fixed seed with a fixed "now" (Thursday 15 October 2026, so Dussehra and Diwali fall inside the booking window): 4 cinemas, 12 screens, 9 films and 1,440 shows with seat maps (41,110 seats taken); 6 bus routes between Hyderabad and Bengaluru, Vijayawada and Chennai with 720 trips (6,966 seats taken); 3 Tandoor Tales branches with 45 tables and 2,613 existing reservations; 200 callers with 510 past bookings (for returning callers) and 70 open bookings. Some shows, trips and slots are sold out on purpose for the "needs an alternative" calls. Each test call starts from a snapshot of the same database.

Gateway and rules. The gateway exposes eleven tools with validated arguments: `search_inventory`, `resolve_date`, `find_alternatives`, `create_booking`, `modify_booking`, `cancel_booking`, `lookup_booking`, `get_caller_history`, `search_policy`, `send_sms` and `handoff_to_staff`. Rules are enforced in code, not in the prompt: at most 10 movie tickets and changes up to 4 h before the show; at most 6 bus seats, name/age/gender for every passenger, refunds of 90/75/50/25% by hours to departure; restaurant party limits (12/10/8 by branch) and a deposit for large groups. A create, change or cancel first returns a

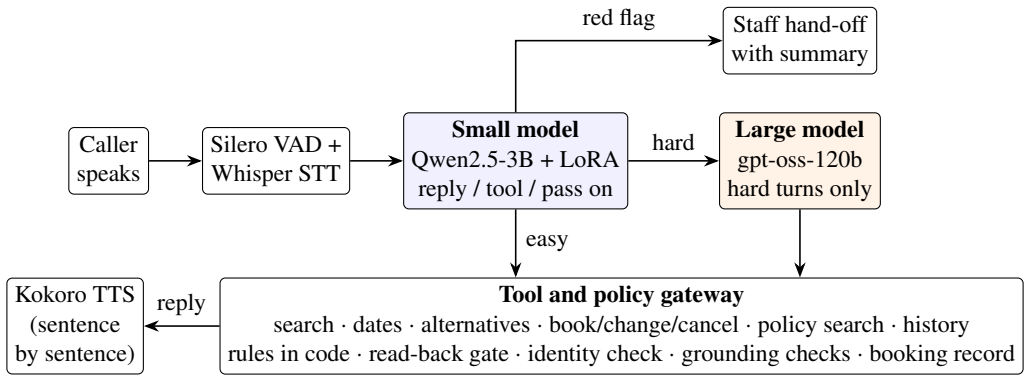


Figure 1: FastLane. Every caller turn goes to the tuned small model, which replies (optionally with a tool call), passes the turn to the large model, or hands it to staff. Neither model writes to the booking system directly.

read-back that must be spoken to the caller; the same call succeeds only after the caller confirms it in the next turn. Changes need the phone number of the booking. Two rules were added after the pilots (Section 7.5): *passenger source* (every bus passenger's name and age must have been said by the caller or come from their history, and the passenger count must cover the seats asked for) and *grounding* (the booked venue and time must have been said by the caller or offered by the agent earlier in the call). There is no payment tool; SMS is only logged.

Booking record and retrieval. A typed per-domain record (film/route/branch, date, time, count, passengers, name, read-back status) is shared by both models and the staff summary. Policy questions use a BM25 index over 25 policy documents (fares, refund rules, FAQs, allergen-tagged menus, a festival calendar, an old and a current version of the bus refund policy); a lookup starts on partial transcripts and is reused if the final transcript matches (early lookup).

Routing. The small model outputs one JSON object: `route` (self / llm / staff), the slots heard this turn, an optional tool call and the words to say. A turn goes to the large model when the small model routes it there or its escalation score (from token log-probabilities) exceeds a threshold, and in code on a tool error, no availability, a changed earlier choice, an unconfident policy match, or (added later) a policy answer containing numbers. It goes to staff on the small model's "staff" route or two failed tool calls in a row. The large model gets the booking record, the last turns and the same tools, finishes the turn, and returns control. Read-back confirmations and goodbyes are templates that need no model call.

Models and serving. Large model: `openai/gpt-oss-120b` on Groq's free tier, with `gemini-3.5-flash` as a backup when Groq would make a call wait more than 10 s (capped at 20 s, no retry); every response records which model answered, and rate-limit waiting is logged separately (`throttle_ms`) and never counted as model time. Simulated caller: Qwen2.5-7B, local. Small models, tuned and untuned, run as 4-bit MLX models (1.6 GB) on a 16 GB Apple M4 behind an OpenAI-compatible server that returns log-probabilities, so both are timed on the same stack.

4 Datasets

Table 2 lists every dataset used so far and the ones still planned. All dialogue data is generated in our practice world, so every call has an exact gold outcome; the public speech data (Svarah) is used only for choosing and testing the recogniser.

5 Experimental Setup: Baselines, Ablations and Metrics

Baselines and ablations. One configuration object defines all 12 systems of the proposal (Table 3). Baselines (B1–B3) are other ways someone could build the same agent; ablations (A1–A8) are FastLane with exactly one part removed or changed, so each differs from FastLane in one flag. All 12 run end to end. Experiments run so far (Section 7.6): FastLane, B2, A2 and A3 on the same 24 test calls (one per domain $\times$ category cell), A8 on the 8 restaurant calls of that sample, and A1 (no fine-tuning), which is the

Dataset	Size	Use and source
Practice booking world	1,440 shows, 720 bus trips, 45 tables, 200 callers, 510 past bookings	SQLite, generated from a fixed seed; every call starts from the same snapshot
360-call test set	120 per domain, 8 categories	scripted callers with gold final states and marked hard turns; never used for training
Training data (generated)	645 calls; 3,153–3,722 train / 245–273 val steps per version	oracle agent run through the real orchestrator; five versions (Table 4); a movies+buses subset for A8
Policy corpus + QA set	25 documents (≈21 pages); 67 questions	written for the world (fares, refunds, FAQs, menus, festivals, old/new policy); retrieval evaluation
Svarah (AI4Bharat)	300 random test clips	real Indian-English speakers, 15+ first languages; STT comparison
TTS caller-line set	142 clips	pilot caller lines spoken by 4 Kokoro voices; key-detail accuracy of STT
Round-trip noise set	3,524 clips (1,762 lines $\times$ 2)	caller lines through TTS, a phone channel and Whisper; noise copies in data v5 and a noisy validation set
Schema-Guided Dialogue, Taskmaster-1	planned (≈2,000 turns)	public booking dialogues (movies, buses, restaurants); not yet converted
Own recordings	planned (50 clips, 20 calls)	real team voices for STT and to check the simulated caller

Table 2: Datasets used so far (top) and planned (bottom).

System	How it differs from FastLane
FastLane	tuned small model on every turn; hard turns to the large model or staff; code tools, gateway, record, early lookup, caller history
B1	large model only, booking rules in the prompt (how most voice agents work)
B2	untuned small model only; large model never used
B3	the large model is asked before every turn whether it is easy or hard
A1–A8	no fine-tuning (A1); no date/alternative tools (A2); rules only in the prompt, no read-back (A3); no booking record (A4); no step-by-step reasoning for the large model (A5); no early lookup (A6); no caller history (A7); small model trained on movies+buses only, tested on restaurants (A8)

Table 3: The 12 systems of the proposal, all implemented as presets of one configuration.

first pilot with the untuned model. B1 and B3 were not run because the daily free quota of the large model ran out; they are the first item of the timeline. Offline ablations on held-out steps cover A1 (untuned vs. tuned) and A8 (new domain), and a data ablation (rule-based vs. round-trip speech noise, data v4 vs. v5) is prepared.

Test set. 360 scripted calls, 120 per domain, never used for training: simple book/change/cancel (120), relative dates and festivals (45), sold out (45), policy questions (40), returning callers (40), change of mind (30), red flags and out-of-scope requests (25), and rule-breaking requests such as waiving a deposit or cancelling someone else's booking (15). Each script gives the persona, goal, opening line, details revealed only when asked, marked hard moments, setup and the gold final database state; every gold outcome was checked to be achievable. 10% of calls get an injected tool failure. An LLM plays the caller in text.

Metrics. Success = the set of bookings created or changed matches one acceptable gold outcome exactly (plus hand-off and policy-answer expectations), with Wilson 95% intervals. Speed = model time per turn (p50/p95), excluding rate-limit waiting. Also: share of turns sent to the large model, large-model calls and tokens per call, router precision/recall on labelled hard turns, invalid tool calls, success on calls with injected errors, unsafe, unconfirmed and rule-violating writes, missed hand-offs, and early-lookup

Data	Train	Val	What changed and why
v1	3,722	273	first version; 3,095 self / 411 llm / 216 staff in train; rule-based noise copies
v2	3,671	267	spoken dates go in a slot and are resolved in code (v1 errors were all dates); A2 escalates every spoken date; "Friday the 23rd" fixed; 516 llm targets
v3	3,671	267	A2 only: after the large model says a date back, the small model stores it as ISO (v2 labels were unlearnable)
v4	3,337	245	callers give 0–5 details in any order, 35% without the domain word (voice demo dropped details)
v5	3,153	245	v4 calls; noise copies from a TTS→STT round trip with a key-detail filter instead of rules

Table 4: Training data versions (all-domains set; steps). Every version keeps 645 of 650 generated calls.

Run	Data	Opt. steps	Time	Val loss	Notes
v1 (run 1)	v1	–	–	–	failed at step 1: bfloat16 LoRA weights on a T4 (fix: cast trainable weights to float32)
v1 (run 2)	v1	466	3 h 17 m	0.0128	token acc. 99.31%; merge on Kaggle failed (torchao), merged locally
v2 all-domains	v2	460	3 h 16 m	0.0118	token acc. 99.30%
v2 movies+buses (A8)	v2	332	2 h 20 m	0.0152	2,644 train / 165 val; token acc. 99.27%
v3 all-domains	v3	460	3 h 22 m	0.0105	token acc. 99.39%; current model
v3 repeat	v3	460	3 h 29 m	0.0113	run meant for v4 used the old dataset version; shows run-to-run variation

Table 5: Kaggle training runs (one T4, QLoRA). Validation sets differ between data versions, so losses across rows are not strictly comparable.

statistics.

Offline step check. For model development we replay held-out validation steps (the exact small-model input of a generated call) and compare the output with the target JSON: valid JSON, route accuracy, escalation precision/recall, right tool and exact tool arguments (the arguments that decide the action; dates count as equal when they name the same day). This needs no large model or simulated caller, so it is cheap and deterministic.

6 Training Data and Fine-Tuning

Generator. Instead of an LLM teacher, both sides of each practice call are code: a rule-based caller that follows a plan (domain, kind of call, which details it gives and when) and an oracle agent that knows the ideal next step. The calls run through the *real* orchestrator and gateway, and every small-model call is stored as (exact input, ideal JSON output), so training inputs match inference byte for byte and are free to produce. Only calls that end in the planned final state are kept (645 of 650). Routing targets follow the proposal: changed choices, "my usual" and refused details go to the large model; red flags, deposit waivers and over-limit groups go to staff; no-availability and tool errors escalate in code. Prompts are mixed (FastLane 60%, A2 15%, A4 10%, A6 5%, A7 10%) so one adapter works with every variant's tool list and context. Escalation steps (about 5% of steps) are repeated $3\times$ in training only. About 30% of training steps get a speech-noise copy (Section 7.9). The tuned model gets a 643-token system prompt instead of 1,062 (less prefill time). Data is split by call, 90/10.

Training. QLoRA on a Kaggle T4: 4-bit base, rank 16, α=32, dropout 0.05, all seven projection matrices, learning rate $2 \cdot 10^{-4}$ with cosine decay and 3% warm-up, effective batch 16, 2 epochs, sequence length 2,048, loss on the target JSON only. Adapters are merged and converted to 4-bit MLX locally. Table 4 lists the data versions and Table 5 the training runs; Figure 2 shows that validation loss flattens from about epoch 1.7, so two epochs are enough.

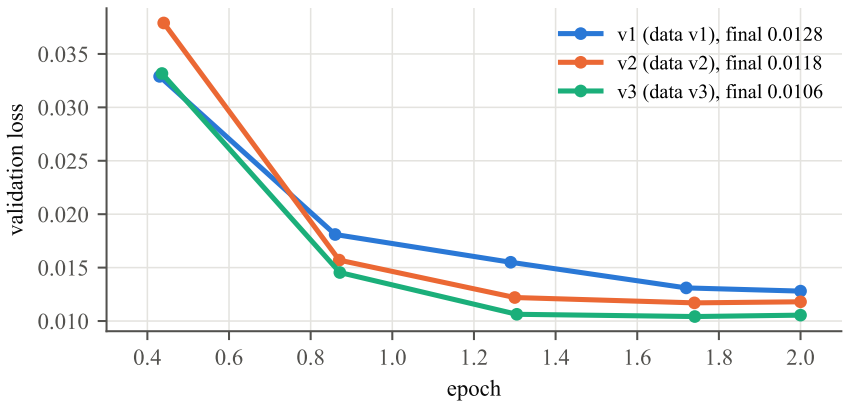


Figure 2: Validation loss of the all-domains adapters by epoch (each on its own data version's validation set).

	267 steps of data v2			267 steps of data v3	
Metric	Untuned	v1	v2	v2	v3
Route accuracy	89.1	96.6	**98.9**	99.3	99.3
Escalation precision	46.2	100	96.0	100	100
Escalation recall	23.1	65.4	**92.3**	92.3	92.3
Right tool	49.8	96.7	96.7	95.9	**98.8**
Exact tool arguments	26.8	90.5	**92.5**	97.2	**99.0**
p50 / p95 latency (s)	2.20 / 7.55	2.22 / 4.83	2.22 / 4.51	2.32 / 4.88	1.96 / 4.24

Table 6: Offline step accuracy (%) of the small model, same steps and serving stack within each block.

7 Results

7.1 Retrieval

On a 67-question policy QA set, BM25 retrieval reaches Recall@3 0.967, top-1 0.817, answer-in-context 0.883, covered when answerable 0.933, current-policy choice 1.00 (old vs. new refund policy) and "not covered" detection 0.714 (only 7 such questions). The thresholds were tuned on the same set, so these numbers are optimistic.

7.2 Offline step accuracy

Table 6 compares the untuned base with the tuned versions on the same 267 held-out steps and the same 4-bit MLX stack. Fine-tuning is what makes the small model usable: the untuned model routes 89.1% of steps correctly but escalates with 46% precision and fills tool arguments exactly only 26.8% of the time. v1 already routed 96.6% correctly; every one of its remaining errors involved dates (a date said early in the call had scrolled out of the 6-line context and was never stored, A2 date turns were not escalated, and `resolve_date` read "Friday the 23rd" as the coming Friday: 69 of 360 spoken forms wrong). Data v2 fixed these in the data and the tools, raising escalation recall from 65.4% to 92.3% and arguments from 90.5% to 92.5% (98.4% in the FastLane configuration). v3 removed a label bug in A2 (Section 8): on its own validation set, steps with any error fell from 15 (v2) to 6, and the date the large model says back is carried into the record in 7 of 10 A2 calls (v2: 0 of 10).

Error analysis of v3. The 6 remaining errors: two changes that call `modify_booking` without searching first, one `quantity` instead of `party_size` in a restaurant change, one extra policy search, one complaint handed to staff by calling the tool instead of routing (same outcome) and one missed A2 escalation.

7.3 New domain (RQ3, offline)

The movies+buses adapter (A8) has never seen a restaurant call. On the 54 restaurant steps of data v2 it routes 98.1% correctly (all-domains v2: 100%; untuned: 87.0%), with escalation recall 88.9% (100%;

24 calls	Pilot v0 (untuned)	Pilot v3 (tuned)
Task success [95% CI]	79.2% [59.5–90.8]	**87.5%** [69.0–95.7]
Turns sent to the large model	60.7%	**13.5%**
Large-model calls / tokens per call (in / out)	6.12 / 12,013 / 1,051	**1.04 / 2,316 / 200**
Rate-limit wait per call	110.3 s	**0 s**
Model time per turn p50 / p95	2.88 s / **6.89 s**	**1.66 s** / 8.06 s
Router precision / hard-turn recall	25.3% / **90.5%**	**76.5%** / 56.5%
Invalid tool calls	7.6%	**0.9%**
Success with injected tool errors	100%	100%
Unsafe / unconfirmed / rule-breaking writes	0 / 0 / 0	0 / 0 / 0
Missed hand-offs	0/3	0/3
Success by domain (buses / movies / restaurants)	6/8, 7/8, 6/8	5/8, 8/8, 8/8

Table 7: 24-call pilots with FastLane; only the small model differs (untuned Qwen2.5-3B vs. adapter v3).

33.3%) and the right tool 88.9% of the time (95.6%; 46.7%). Argument filling does *not* transfer: 47.6% exact (100%; 50.0%), and all 11 errors are the same one, `quantity` (the movie/bus field) where restaurants need `party_size`. The gateway rejects such calls, so in a live call this costs an extra step or a hand-off. Routing transfers; the schema does not.

7.4 Pilot: untuned vs. tuned small model

Table 7 compares two 24-call pilots (one call per domain × category cell, same calls, same caller simulator and large model); only the small model changed. The tuned model keeps easy turns: the share of turns sent to the large model falls from 60.7% to 13.5%, large-model tokens per call fall 5×, the free-tier rate limit is never hit (110 s of waiting per call → 0), and median model time per turn falls by 42%. Success rises from 79.2% to 87.5% (policy 0/3 → 2/3, sold out and change of mind 2/3 → 3/3). Two numbers got worse. *p95* rose because the slowest turns are now booking turns where the small model takes two steps in a row (search, then read-back: about 3 s + 5 s on MLX), plus one 45.6 s Gemini reply. *Router recall* fell because the test labels mark every relative date as hard, while FastLane handles dates on the small model with code tools by design; 5 of the 13 "missed" hard turns are such date turns, and those calls succeeded. The intervals overlap (24 calls, one seed): this is a pilot, not the final comparison.

7.5 Failures and the fixes they led to

All three failed calls of pilot v3 were buses, and all were the small model acting alone where it should not (Table 8). No gateway rule was broken, but two calls showed that passenger details were not checked against what the caller had actually said. The voice demo later showed a related gap (a search with a cinema nobody named and without the time the caller gave), which led to the grounding rule. Both rules were checked against all 650 generated practice calls: the same 645 calls are kept and *no* training example changes, so the rules never block a correct booking in our data. After the fixes, the 3 failed calls were rerun and all 3 succeed.

7.6 Baselines and ablations on 24 calls

For the mid-submission we ran the baselines and ablations on the same 24 calls as the pilots, with adapter v3, the fixes above, the same caller simulator and the same large model (Table 9, Figure 3). Each run is one seed; the intervals are wide, so only large gaps are meaningful.

FastLane. The fixes of Section 7.5 lift success from 87.5% to 95.8% (23/24); buses 8/8, movies 8/8, restaurants 7/8. It sends more turns to the large model than before the fixes (21.3% vs. 13.5%) because numeric policy answers and refused bookings now go there by design, but it still has the lowest median model time of all systems. The one failure, res-policy-321, gave the correct deposit (Rs 2,400), but the simulated caller's next reply was read as a change of mind and the call went to staff. There were no unsafe writes and no missed hand-offs, and all calls with an injected tool error succeeded.

Call	What went wrong	Fix and outcome on rerun
bus-dates-161	asked “Which cinema?” in a bus call, then booked invented passengers (“Caller (40)”)	passenger-source rule: the booking is refused, the large model asks for both passengers; booked correctly
bus-rule_break-236	caller asked for 3 seats but gave 1 passenger; model booked 1 seat with an invented age	passenger-count check: two attempts refused, call handed to staff (an accepted outcome)
bus-policy-201	small model misread the refund table (90% instead of 25% for 10 h before departure)	policy answers with numbers go to the large model: answered 25%
mov-sold_out-056	one Gemini backup reply took 45.6 s	backup capped at 20 s, no retry, lost time counted as waiting

Table 8: Pilot v3 failures, the code fixes (no retraining) and the rerun.

System	Calls	Success [95% CI]	Large-model turns	Large calls per call	p50 / p95 (s)	Invalid tool calls	Unsafe-write calls	Router P / R
FastLane	24	**95.8%** [80–99]	21.3%	1.67	**1.55** / 7.59	**0.8%**	**0**	**65** / 77
FastLane, before fixes	24	87.5% [69–96]	13.5%	1.04	1.66 / 8.06	0.9%	0	77 / 57
B1 large only	–	not run: daily API quota exhausted (first item of the timeline)						
B2 untuned small	24	29.2% [15–49]	0%	0	2.12 / 19.74	26.1%	0	–
B3 large as router	–	not run: daily API quota exhausted						
A1 no fine-tuning†	24	79.2% [60–91]	60.7%	6.12	2.88 / **6.89**	7.6%	0	25 / **91**
A2 no date tools	24	87.5% [69–96]	43.5%	3.58	2.94 / 7.76	2.8%	0	35 / 87
A3 rules in prompt	24	75.0% [55–88]	24.1%	1.88	2.20 / 9.82	3.6%	**15**	38 / 58
A8 movies+buses	8‡	75.0% [41–93]	25.0%	1.62	1.52 / 5.49	12.1%	0	33 / 60

Table 9: Baselines and ablations on the 24 pilot calls (one seed). Model time excludes rate-limit waiting. †Pilot v0, run before the gateway fixes. ‡Restaurant calls only; FastLane succeeds on 7 of the same 8.

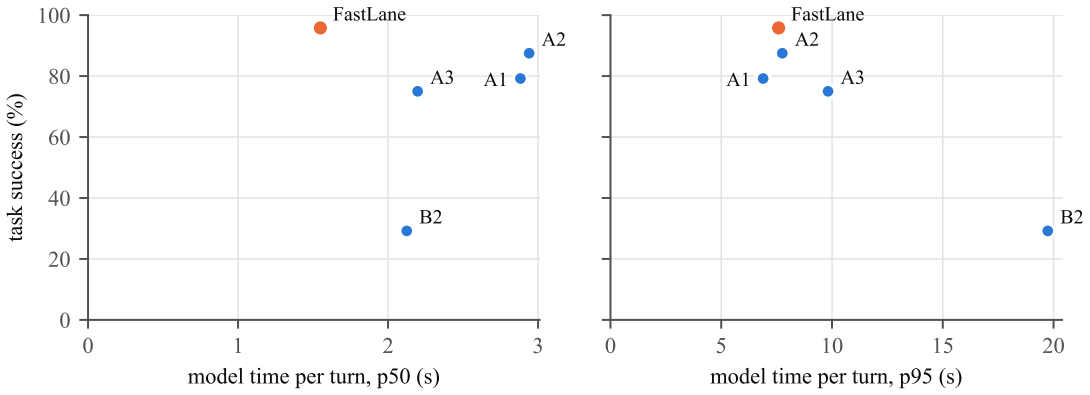


Figure 3: Task success vs. median (left) and p95 (right) model time per turn on the 24 calls. FastLane has the highest success and the lowest median; A1 has a lower p95. B1 and B3 are missing (API quota).

B2: the untuned small model alone fails. It succeeds only where the right outcome is a refusal or a hand-off (red flags 3/3, rule-breaking requests 3/3, returning callers 1/3, every other category 0/3). A quarter of its tool calls are invalid, it loops through many steps (34.5k prompt tokens per call vs. 3.7k for FastLane, 226 turns vs. 108), and its p95 is 19.7 s. A small model can only replace the large model on easy turns once it is tuned.

A1: no fine-tuning. The untuned small model in FastLane (pilot v0) passes 60.7% of turns on and makes 6.1 large-model calls per call (FastLane: 1.7), with 79.2% success. Its p95 is the lowest of all systems because the large model answers most turns in one step; the cost is 3.7× the large-model calls and a 110 s rate-limit wait per call on the free tier. A1 also predates the gateway fixes, so it differs from FastLane in more than one flag.

A2: no date and alternative tools (H4). Without `resolve_date` and `find_alternatives` the small model cannot handle dates or sold-out shows, so it passes them on: the share of turns sent to the large model doubles (21.3% → 43.5%), large-model calls per call rise 2.1× (1.67 → 3.58), large-model tokens per call rise 1.5×, and median model time per turn rises from 1.55 s to 2.94 s. Success falls from 23/24 to 21/24: one wrong date in a change of mind (15 October instead of 23 October), one booking that was never made, and one policy question handed to staff. A2's router recall looks better (87% vs. 77%) only because it passes almost everything on.

A3: rules only in the prompt, no read-back (H5). This is the clearest result. With the same models but no gateway rules and no read-back gate, **15 of 24 calls made a booking change that the caller never confirmed** (18 such writes) and 4 writes broke a booking rule; FastLane made none. All 3 rule-breaking calls failed: two bookings were changed although the rules forbid it, and 1 seat was booked for a caller who asked for 3 passengers. Ten of the 15 calls with unconfirmed writes still ended in the right state, so task success alone (75%) understates the risk: the rules and the read-back are what make the writes safe.

A8: new domain end to end (H6). The movies+buses adapter succeeds on 6 of the 8 restaurant calls (FastLane 7/8). Half of the calls (4/8) contained one invalid tool call (12.1% of tool calls vs. 0.8%), the `quantity`/`party_size` mismatch found offline; the gateway rejected each one and 3 of these 4 calls still succeeded. The extra failure was a change of mind where the old time (14:00) was kept instead of the new one (12:30). A8 uses the v2 adapter (the movies+buses adapter was not retrained on data v3), so the gap mixes domain and data version.

7.7 Voice pipeline

`fastlane voice` serves a browser page; one WebSocket per call carries 16 kHz audio. Silero VAD ends a turn after 600 ms of silence and sends partial transcripts every 0.6 s to early lookup; Whisper large-v3-turbo on MLX transcribes with a domain word list (film, cinema, branch, city and festival names); the unchanged FastLane agent replies; Kokoro speaks the reply sentence by sentence. It is half duplex, with a typed-text fallback. We used a small FastAPI loop instead of Pipecat so that every stage is timed in one trace. In an 8-turn spoken booking (TTS-made caller) the median times were: STT 1.04 s, agent 1.3 s, first reply audio 0.45 s, and **3.2 s from the end of speech to the first reply audio** (plus the 0.6 s silence wait). The slowest turn (10.2 s) was a read-back: two small-model steps (5.8 s) and a long first sentence; a short lead-in ("Just to confirm:") is now spoken as its own chunk. Live calls exposed three hand-off problems, now fixed in code: the large model said it would transfer the caller but called no tool (the hand-off is now made in the same turn when a reply announces it), Markdown was read aloud (replies are converted to plain sentences and lists), and the page hid the end of the call.

7.8 Speech recognition on Indian English

Table 10 compares recognisers on 300 random clips of the Svarah test split (real speakers with 15+ first languages, general topics) and on 142 caller lines from the pilots spoken by four Kokoro voices, two of them Hindi-accented. Key-detail accuracy is the share of places, films, names and numbers in the reference that appear in the transcript after normalisation. Sarvam is best on real Indian speech by a wide margin and 3× faster (best or near-best for every first-language group, e.g. Kannada 7.0% vs. 11–43%).

	Svarah (300 clips)		Booking lines, TTS (142 clips)			
Recogniser	WER	s / clip	WER	Places	Names	Numbers
Sarvam saarika:v2.5 (cloud)	**5.8**	**0.46**	8.1	80.6	**36.8**	96.7
Whisper large-v3-turbo (MLX)	10.5	1.42	9.4	48.4	26.3	95.8
+ domain word list	14.9	1.40	8.0	**96.8**	26.3	94.2
faster-whisper small.en (CPU)	9.5	1.34	10.8	48.4	26.3	**97.5**
+ domain word list	12.3	1.50	**7.8**	**96.8**	26.3	**97.5**

Table 10: Speech recognition: WER (%), median time per clip, and key-detail accuracy (%) on booking lines.

The domain word list helps only in domain: it lifts places on booking lines from 48% to 97% but raises WER on Svarah's general speech by 3–4 points. Person names are the weak spot for every recogniser (26–37%). We also found and fixed a setting of ours: Whisper had been run with temperature fixed at 0, which disables its fallback decoding and let short clips fall into repetition loops (turbo + word list: 20.1% WER before, 14.9% after).

7.9 Speech noise in training data

Since data v1, about 30% of training steps have a copy whose caller lines are rewritten by hand-made rules ("Kukatpally" → "kuka pally", "four" → "for") with the same target. Measured on the 1,762 caller lines of data v4, **14.9% of these rule-based copies lose a key detail that the target still contains** (e.g. "for tickets" with target `quantity: 4`), which teaches the model to fill in values nobody said; a live call did exactly that (a misheard "Authenticates" read back as 4 tickets). Data v5 replaces the rules with a round trip: every caller line, twice, is spoken by a Kokoro voice (half Hindi-accented, speed 0.9–1.15), passed through a phone channel (300–3400 Hz, noise at 12–30 dB SNR) and transcribed by the same Whisper set-up the voice demo uses. A transcript is used only if every key detail survives (places, films, dates, names, numbers, booking references, short forms such as "Oct"/"Hyd", and yes/no polarity, since "Haan . . . na" heard as "maybe not" must not become a yes). Of 3,524 clips (mean WER 35.7%, median 20%), 55.7% are usable; names are lost most often (1,334 clips), then numbers (335).

To see how much recogniser noise hurts the current model, we ran v3 on the 245 validation steps of data v4 once with clean caller lines and once with usable round-trip transcripts (192 steps changed). Routing is unaffected (98.8% vs. 98.8%; escalation recall 88.5% both; right tool 96.3% vs. 96.8%), but exact arguments fall from **98.1% to 94.3%**. The new errors are the kind the round-trip copies target: twice "8:30 PM" was transcribed as "8th, 30 p.m." and v3 searched for 18:00, a time nobody said; a party of 5 was searched as 2; "on the bus?" became "on the back?" and a policy search went to restaurants. Data v4 and v5 contain the same calls and differ only in the noise copies, which makes them a clean ablation once both are trained.

8 Challenges Faced and Mitigation

Challenges already solved. Every retraining and fix so far was driven by an observed failure; Table 12 summarises the chain. Three lessons generalise. (1) *Most model errors came from the data, not from training*: every v1 error traced back to how dates were represented, and v2's A2 errors to labels that could not be learned (the date the large model said back was never written into the record, so after it scrolled out of the context the target date was invisible). (2) *Check that a new model is really new*: the first v2 evaluation reproduced v1 exactly because a stale merged folder was reused; the merge now always starts from scratch and every build is checked by weight hash. The same lesson recurred on Kaggle, where a run meant for v4 trained the old dataset version (Table 5). (3) *Safety belongs in code*: no FastLane run produced an unsafe write, when the small model invented passengers or a cinema a gateway rule stopped it without retraining, and the A3 ablation shows what happens without these rules (15 of 24 calls with unconfirmed writes).

Open challenges and mitigation plans. Table 13 lists the challenges that still affect the final results and how we plan to handle them before 15 November.

Hypothesis	Expected	Evidence so far
H1 Routing is faster	p95 $\geq$50% lower than B1	B1 not yet run (quota). FastLane has the lowest p50 of all systems run (1.55 s); its p95 (7.6 s) is above A1's (6.9 s), so the p95 goal is open
H2 Quality holds	within 3 points of B1	B1 not yet run; FastLane 95.8% [80–99], the highest of the systems run (24 calls)
H3 Router catches hard turns	$\geq$90% passed on	offline escalation recall 92.3% (v2/v3); pilot 56.5% under a label definition that counts date turns as hard
H4 Tools cut large-model use	clearly lower with tools	**supported**: without the tools (A2) large-model turns 21.3% $\rightarrow$ 43.5%, calls per call 1.67 $\rightarrow$ 3.58, success 23/24 $\rightarrow$ 21/24
H5 Rules in code are safer	fewer unsafe writes	**supported**: rules in the prompt only (A3) give 15/24 calls with unconfirmed writes and 4 rule-breaking writes; FastLane 0
H6 Routing transfers	within 5 points on restaurants	offline routing 98.1% (vs. 100%), arguments 47.6% (field mismatch); end to end 6/8 vs. 7/8 restaurant calls (one call = 12.5 points)
H7 Early lookup hides delay	lower p95 on lookup turns	not shown: in the 24-call run 9 lookups started, 2 reused, 3 wasted, negligible time saved

Table 11: Hypotheses of the proposal and the evidence so far.

Where	Challenge	Mitigation (done)
Pilot v0	untuned model sends 61% of turns to the large model; refund-rule questions go to staff	fine-tune (data v1, adapter v1)
Adapter v1	all remaining errors involve dates (context loss, A2, weekday bug)	data v2: dates as slots resolved in code; `resolve_date` fix
Adapter v2	first check identical to v1	stale merge folder; merge from scratch, hash check
Adapter v2	A2 date labels unlearnable; 5 wrong A2 dates	data v3 and adapter v3 (A2 carries the ISO date)
Scoring	"October 24" scored wrong against 2026-10-24	dates compared by day; saved rows rescored
Pilot v3	invented passengers, partial booking, misread refund table	passenger-source rule, numeric policy answers to the large model
Voice demo	details dropped from long openings; unrequested cinema	data v4 (multi-detail openings); grounding rule
STT on Svarah	repetition loops on short clips	restore Whisper's temperature fallback
Rule noise	14.9% of noise copies contradict their target	data v5: TTS$\rightarrow$STT round trip with a key-detail filter
Live calls	announced but unmade hand-offs; Markdown read aloud	same-turn hand-off; plain-speech replies
Mid runs	Groq daily token cap (200k) and the Gemini backup's free tier ran out during the runs	resumable runner kept every finished call; waits excluded from model time; A2 finished overnight; B1 and B3 moved to the timeline

Table 12: Challenges met so far and how each was resolved (details and evidence files are in the run log).

Challenge	Impact	Mitigation plan
Large-model rate limits (Groq free tier: 8k tokens/min, 1k requests/day; one call carries ≈3k tokens)	B1 and B3 could not be run for this report; full plan (≈13,000 calls) impossible on the free tier; backup model (Gemini, 4–7 s per call) mixes latencies	scaled-down Phase 8 (below); waiting logged apart from model time; report which model answered; a paid tier for B1 and FastLane if available
Two weeks to the final submission	12 systems × 360 calls × 3 seeds will not fit	full 360 calls for FastLane, B1, B2; a stratified 120-call subset for B3 and A1–A8; 3 seeds only for FastLane and B1
p95 latency (two small-model steps per booking turn)	H1 depends on p95	merge search and read-back into one step; template read-backs
Router-recall metric counts date turns as hard	understates FastLane's recall	recompute with FastLane's own hard-turn definition (dates resolved in code)
Names misheard by every recogniser (26–37% exact)	wrong passengers, failed bookings in voice calls	spell names back before booking; read-back gate already blocks silent errors
New-domain arguments (A8: `quantity` vs. `party_size`)	A8 calls fail on the schema, not on routing	field alias in the gateway, reported separately from routing
Simulated callers may be unrealistic	success rates may not transfer to real callers	20 real team calls and 30 transcripts checked against simulated ones
Kaggle runs used a stale dataset version once	lost a 3.5 h run	check data sizes in the notebook before leaving it to run; weight hashes for every build

Table 13: Open challenges and mitigation plans for the final submission.

9 Status of the Hypotheses

Table 11 states what the evidence so far says about each hypothesis. The ablations support H4 (code tools for dates and alternatives halve the large-model share) and H5 (rules and read-back in code remove unconfirmed and rule-breaking writes). The main comparison against B1 (large model only) could not be run before this report, so H1 and H2 are open; on the systems run so far, FastLane has the highest success and the lowest median model time, but not the lowest p95. H6 holds for routing but not for tool arguments, and H7 has no support yet. All of this rests on 24 calls with one seed; the final runs use 120–360 calls and three seeds for the main systems.

10 Remaining Work and Timeline to the Final Submission

The final submission is due on 15 November 2026, about two weeks after this report. Table 14 gives the plan day by day. The order puts first what decides the headline result (RQ1, H1/H2): the next model version, the fixes found so far, and the main comparison against B1.

If time runs short. The conversion of Schema-Guided Dialogue and Taskmaster-1 into training data is the first thing we would drop: the generated data already gives exact tool calls and gold states, and the public data would mainly add language variety. Next, the 120-call subset for the ablations would shrink to 72 calls (one per domain × category × 3), and seeds would drop to one. The comparison of FastLane with B1 on all 360 calls stays in every version of the plan.

11 Limitations and Safety

All results so far come from synthetic callers, a practice booking system and at most 24 end-to-end calls with one seed; apart from B2, the confidence intervals of all systems overlap, and B1 and B3 are still

Dates	Work	Output
30 Oct	mid-submission; FastLane, B2, A2, A3, A8 on 24 calls	this report
31 Oct–1 Nov	B1 and B3 on the same 24 calls as the daily quota resets	complete Table 9; first H1/H2 check
31 Oct–2 Nov (in parallel)	train data v5 on Kaggle (both adapters, ≈5 h); offline check of v5 vs. v3 on the clean and round-trip validation sets; code fixes: heard-number check for ticket counts, `party_size` alias (A8), spelling names back, one-step search + read-back (p95), FastLane hard-turn definition for router recall	adapter v5; fixes with tests
3–4 Nov	v4 run for the noise ablation (if Kaggle quota allows); 24-call check of v5 with all fixes	noise-ablation table
5–9 Nov	Phase 8: FastLane, B1, B2 on all 360 calls; B3 and A1–A8 on a stratified 120-call subset; 3 seeds for FastLane and B1	success vs. p95 plot, unsafe-action table, 95% intervals
5–9 Nov (in parallel)	two team members label 300 turns (Cohen's κ); 20 real calls and 50 recordings; policy corpus to ≈60 pages with a held-out QA set	κ, real-call check, retrieval numbers
10–11 Nov	voice runs on 50 calls per main system; Sarvam STT in the demo	end-to-end latency
12–13 Nov	analysis of H1–H7 and RQ1–RQ3; error analysis; figures	results section
14–15 Nov	final report, README with results, demo video; code freeze on GitHub	final submission

Table 14: Timeline from the mid-submission to the final submission (15 November 2026).

missing. The large model is a free-tier model with a backup, so its latency varies. The round-trip noise uses synthetic voices, which are harsher than Svarah's real speech in places and softer in others. On safety, the system follows the proposal's plan: a mock system only, no payment tool, read-back before any change, identity check by phone number, hand-off with a summary for complaints, payments and out-of-scope requests, and an agent that states it is automated. All test callers are synthetic; API keys stay in a local environment file and are never logged.